

Value Proposition Canvas

Products & Services

- What are the products and services you can offer your customer so they can get their job done?
 - The products and services offered include a portable diagnostic device and disposable breath cartridges that enable at-home, non-invasive lung cancer screening. Additional supporting services include technical support, provider training, educational outreach, and integration assistance for healthcare institutions. These offerings help customers, both individual patients and larger groups such as hospitals, clinics, labs, and underserved communities, to perform early lung cancer detection and routine monitoring both efficiently and accessibly.

Gain Creators

- What can you offer your customer to help them achieve their gains?
 - The product creates gains by offering a portable, low-cost, and non-invasive diagnostic tool that enables early detection of lung cancer, which greatly improves treatment outcomes. It also increases accessibility for underserved populations, supports routine monitoring through a reusable device with disposable cartridges, and reduces reliance on traditionally invasive or radiation-based diagnostics. Additionally, it also provides training and support for healthcare providers and builds clinical credibility through partnerships with research institutions and other medical organizations for the present and future.

Pain Relievers

- How can you help your customer to relieve their pains?
 - The product will help relieve our customers' pain by addressing several key barriers in current lung cancer screening practices. Many individuals at high risk do not participate in traditional screening due to its invasiveness, cost, radiation exposure, and limited access to facilities. Our product offers a non-invasive, low-cost, at-home breath test that uses RNA aptamer biosensors to detect lung cancer early through exhaled breath condensate.
- What problems can you eradicate?
 - By making early detection easier and more accessible, we help reduce the emotional, financial, and logistical burden on patients and healthcare providers.

Gains

- What would make your customer happy?

Sundew Dx

Denis Victorov, John Vorwald, Xiujun Fan, Alexandra Jensen, Zaid Slayyeh, Rohan Kumar, Tiya Shah, Ruthie Carmeli

- What would make our customers happy would be for our diagnostic device to be able to successfully detect suspected cancerous VOCs with high accuracy/precision, so the patient may be aware and can get further tests done. This would allow for the possible early detection of developing lung cancer, making survival much more likely and treatment easier and more effective. Another aspect of our device that would make our customers happy would be that it is a much more affordable alternative than the current lung cancer diagnostic technologies, and that would heavily relieve the financial burden that is had when diagnosing lung cancer. Our technology will be easy and simple to use, and the customer will also be very happy, as other diagnostic devices are much more complicated to use with more extensive instructions. Customers would also appreciate how quickly they could get started using our technology. Lastly, our device would make our customers happy as there is no biopsy or surgery needed with our diagnostic device, as it is non-invasive, a much better alternative when compared with traditional diagnostic tools that require a biopsy and other invasive procedures.
- What would make their life and the job-to-be-done easier?
 - The device would be small, easy to handle, simple and easy to use, and would be quick to conduct a diagnostic test, and would give you almost instantaneous results. The UX/UI would also be easy to use and informative! This device is non-invasive, therefore, there would be no recovery time needed and no injury sustained, like with other invasive procedures needed in other diagnostic methods, which can also play as a significant price reduction to getting a diagnostic test, and this will make the patients' experience infinitely more enjoyable. There would be a low price barrier, so it would be easier for our patients to use, as it wouldn't be a big financial burden that they could not afford. The patient wouldn't even have to leave the comfort of their own home as this test could be taken from anywhere, and results could even be sent digitally without ever needing to step foot in the clinic.

Job-to-be-Done

- What is the job the customer wants to get done in their work or life?
 - The job the customer wants to get done using our device would be to see if they have an indication of possible lung disease, and as a result, know if they should get further tests done to get to the next steps to take care of their potential lung cancer. The customer with current technologies has to deal with injuries due to surgeries and recovery time, with invasive diagnostic methods, and this is not something a customer wants to deal with; and would want a test that doesn't have these painful disadvantages. With a non-invasive method, their issues are that they are very expensive, and so a customer would want a much less expensive test to get done, and that is what Sundew's lung cancer diagnostic device can do.

Sundew Dx

Denis Victorov, John Vorwald, Xiujun Fan, Alexandra Jensen, Zaid Slayyeh, Rohan Kumar, Tiya Shah, Ruthie Carmeli

Pains

- What is annoying or troubling your customer?
 - What may be annoying or troubling to customers is having to have to take further steps, going to see the doctor and the process involved with that, and to have more tests done to confirm the results of Sundew's diagnostic results; the thought of this may be troubling for a user and may reduce the likelihood that they even take the test in the first place. Some aspects of the UI/UX may still be annoying or troubling to the user, even after extensive testing and surveying, and to address this, we will welcome feedback directly from the user, through the application itself (will have a recommendation/feedback portion)
- What is preventing them from getting the job done?
 - Even though the Sundew lung cancer diagnostic test takes relatively little effort to use by design, this actually may be enough to feel troubling to the user and deter them from using it. A customers self-worry or carelessness may also prevent them from using the test, as there are some people who may not even want to know they have cancer or not for the sake of their peace of mind, which is why we would also have these devices in the doctors office to be apart of the annual check up people go to.